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REMANENCE IN AUTHIGENIC MAGNETITE: TESTING THE HYDROCARBON-MAGNETITE HYPOTHESIS

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Abstract. Paleomagnetic, rock magnetic, petrographic, and geochemical studies of hydrocarbon-saturated speleothems in southern Oklahoma indicate there is a relation between hydrocarbons and a chemical magnetization that resides in magnetite. The speleothems, which are composed of light and dark calcite bands, occur in caves of karst origin in the Ordovician Kindblade Formation. Vertebrate fossils interbedded with the deposits indicate they are Permian in age. The dark bands contain primary hydrocarbon-filled fluid inclusions. The dark calcites possess over an order of magnitude stronger natural remanent magnetization than the lighter bands which do not yield stable decay during demagnetization. Alternating field (AF) and thermal demagnetization of specimens of the dark bands reveal a Permian direction of magnetization (declination=160, inclination=3, k=17, α95=5, n=56). The results of rock magnetic experiments, and the fact that most maximum unblocking temperatures are below 580°C, suggest that the dominant component resides in magnetite. In some specimens stable decay to 640°C suggests the presence of a weak component residing in hematite. The presence of authigenic magnetite spheres in magnetic extracts of the dark calcites supports a chemical origin for the magnetization. Shallow burial depths probably eliminate the possibility of a thermoviscous magnetization. The occurrence of primary hydrocarbon inclusions suggests that hydrocarbons seeped into the caves during precipitation of the speleothems and were trapped in the calcite crystals. The relationship between intensity of magnetization and hydrocarbon abundance leads us to propose that chemical conditions created by the hydrocarbons caused precipitation of authigenic magnetite and acquisition of the associated chemical remanence.

Introduction

Secondary magnetizations that reside in magnetite and were acquired long after deposition have been reported from several units [e.g., Benthien and Elmore, 1987; McCabe et al., 1983, 1984; Wisniowiecki et al., 1983]. Authigenic magnetite has been reported from these units, and the magnetizations are interpreted as chemical (i.e., diagenetic) in origin. A chemical origin for these magnetizations, however, is controversial from several viewpoints. For example, as demonstrated by Kent [1985], a thermoviscous origin caused by moderately elevated burial temperatures is also a permissible interpretation for some secondary magnetizations residing in authigenic magnetite.

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A second problem is that the mechanism for the diagenetic origin of the authigenic magnetite is not understood.

Several different geochemical mechanisms have been proposed for magnetite authigenesis [e.g., Elmore et al., 1987; McCabe et al., 1987; Ellwood et al., 1988]. One mechanism that has received considerable recent attention is based on the presence of hydrocarbons. For example, Donovan et al. [1979] reported magnetite in altered Permian clastic rocks above the producing horizons at the Cement oil field, Oklahoma and hypothesized that hydrocarbon-related brines migrated from depth upward along faults, causing the reduction of hematite and iron hydroxides and the precipitation of magnetite. Reynolds et al. [1985] in their study of Cement field suggested that the magnetite reported by Donovan et al. [1979] represents contamination from drilling, but did report ferrimagnetic pyrrhotite and suggested that it precipitated as a result of hydrocarbon-related fluids. Although the results from Cement are controversial, recent studies have demonstrated that there is a relationship between hydrocarbons and authigenic magnetite. For example, Elmore et al. [1987] and McCabe et al. [1987] report the presence of authigenic magnetite in samples of solid bitumen that is biodegraded. Elmore et al. [1987] also present preliminary evidence that hydrocarbons can cause acquisition of a stable remanence in authigenic magnetite. In this paper the detailed evidence for a genetic relationship between hydrocarbons and acquisition of a chemical remanence is presented.

The principal evidence for a genetic relationship between hydrocarbons and acquisition of a chemical remanent magnetization (CRM) comes from the study of hydrocarbon-bearing, late Paleozoic speleothems which are found in caves of karst origin in the Lower Ordovician Arbuckle Group in southern Oklahoma. Additional support for the relationship comes from the study of the hydrocarbon-bearing host limestones.

Geologic Setting

Samples for this study were collected from one locality (BQ, Figure 1) in the Limestone Hills, which are north of the Wichita Mountains in southern Oklahoma. Calcite speleothem deposits, some of which are horizontal and are paleodip indicators, were extensively sampled at this locality where they are well exposed. Samples were also collected from the host carbonate unit, the Ordovician Kindblade Formation (Upper Arbuckle Group) which dips at 26° to the northeast (N40E).

The Limestone Hills are on the southern margin of the Southern Oklahoma Aulacogen. They are bounded on the north by the Mountain View fault and the Anadarko Basin and on the south by the Meer's fault and the Wichita Mountains.

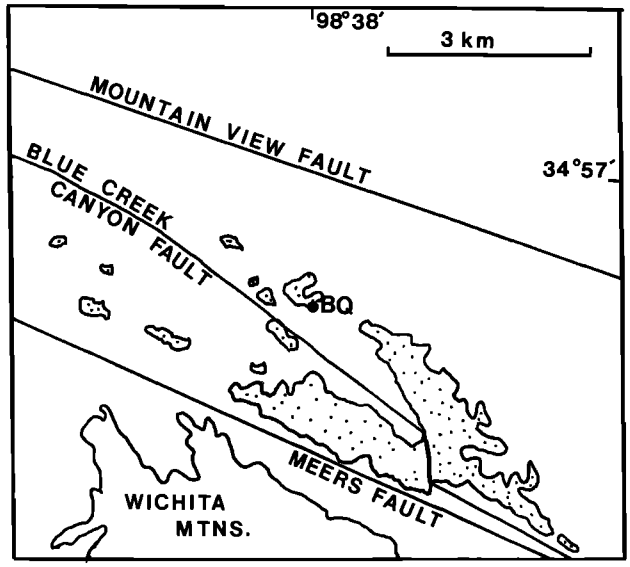


Fig. 1. Map of the Limestone Hills area in southern Oklahoma with the sampling location.

Deformation in the Southern Oklahoma Aulacogen is late Paleozoic in age and is characterized by several structural styles [Donovan, 1986]. For example, reverse faults and tight folding indicate compression [Brewer et al., 1983]. Left-lateral wrench movement along faults parallel to the trend of the aulacogen was also important. Wickham [1978] cites the trends of the folds and faults, and other structural evidence from the region, in support of a wrench fault model. Donovan [1986] suggests that a left-lateral transpression model should be applied to the area, and McCoss and Donovan [1986] present fold and shear array data supporting the model. Elmore et al. [1988] present paleomagnetic evidence from the Limestone Hills that is consistent with left-lateral wrench faulting during deformation of the aulacogen.

Previous paleomagnetic studies of hydrocarbon-bearing Arbuckle Group units in the Limestone Hills suggest that the rocks contain a postfolding late Pennsylvanian-Permian magnetization residing in magnetite [Elmore et al., 1988]. Studies of similar Arbuckle Group carbonates which do not contain hydrocarbons in the Arbuckle Mountains (120 km east) indicate that the carbonates possess late Pennsylvanian-Permian CRMs that reside in hematite [Steiner, 1973; Elmore et al., 1985; Cochran and Elmore, 1987]. These CRMs in hematite have been related to specific diagenetic processes such as dedolomitization [Elmore et al., 1985] and formation of Liesegang bands [Cochran and Elmore, 1987].

Laboratory Methods

Paleomagnetic and Rock Magnetic Samples

Samples were collected with a portable drill and oriented with a clinometer and compass. Specimens (cylinders 2.2 cm in length and 2.5 cm in diameter) were cut from the samples, and their natural remanent magnetizations (NRMs) were measured in a United Scientific three-axis cryogenic magnetometer. Samples were measured and demagnetized in a laboratory which is shielded (<250 nT) from the Earth's magnetic field. Specimens were demagnetized using both alternating field (AF) and thermal demagnetization techniques. During AF treatment specimens were demagnetized in 10 mT steps up to 150 mT in a shielded two-axis tumbling unit. Other specimens were heated in a stepwise manner using a Schonstedt TSD-1 unit (13-15 steps from 100° to 700°C) for 30 min at each temperature and cooled in a field of no more than 2 nT. The direction and intensity of magnetization results were plotted as orthogonal projections of the vector endpoints of each demagnetization step [Zijderveld, 1967]. The direction of magnetization removed from each specimen was calculated using principle component analysis [Kirschvink, 1980].

Representative specimens from each site were chosen for isothermal remanent magnetization (IRM) acquisition and decay experiments to aid in the identification of magnetic mineralogy. Acquisition of IRM was accomplished by subjecting the specimens to an external field up to 0.8 T in a stepwise fashion using a water-cooled electromagnet. Thermal decay of IRM was conducted in 50° steps up to 700°C. Some specimens were subjected to an IRM in the same stepwise manner to 0.8 T, and then an antiparallel IRM was given in a field of 0.2 T. Thermal decay of this IRM was conducted in 100°C steps to 400°C and then in 20°C steps to 700°C.

Petrographic Samples

Outcrop samples and chips cut from cores were used for petrographic analysis. Thin sections were analyzed using both reflected and transmitted light to determine mineralogy and identify diagenetic features. Samples and chips were etched in 10% glacial acetic acid and 10% hydrochloric acid and analyzed using scanning electron microscopy-energy dispersive analysis (SEM-EDA). Secondary electron and backscattered images were viewed.

Extraction procedures were used to separate magnetic and nonmagnetic phases. Extreme care was taken during collection and subsequent extraction procedures to reduce the likelihood of contamination. The speleothems and limestones were dissolved in 1.5-M glacial acetic acid buffered to pH=4 with sodium acetate. The residue was run through the magnetic extraction line. Extracts were examined using X-ray diffraction and SEM-EDA techniques.

Geochemical Samples

Organic material isolated from dark bands of the speleothems was analyzed for the bands' respective n-alkane compositions [Elmore et al., 1987]. The saturate fractions of the dichloromethane-soluble portions of the samples were isolated using gas chromatography (GC) to determine their respective C_{15+} n-alkane distributions. In addition, the CH_2Cl_2 extract of the calcite speleothem was analyzed for its stable carbon isotopic composition. Finally,

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portions of the light and dark calcite bands of the speleothem were ground to fine powders (200 mesh), exhaustively extracted with CH_2Cl_2, and boiled in H_2O_2 to remove organic matter. Portions of the purified calcite were subsequently digested with phosphoric acid and analyzed for their respective stable carbon and oxygen isotopic compositions.

Results

Speleothems: Bally Quarry

The speleothems occur as flowstones and were collected from caves and fissures in Arbuckle Group limestones at location BQ (Figures 1 and 2). Amphibian and reptilian fossils found interbedded with the flowstones indicate that the deposits are Permian in age (L. Jacobs, Southern Methodist University, personal communication, 1986). The speleothems are primarily composed of clear to dark calcite bands, with the dark bands containing abundant primary hydrocarbon-filled fluid inclusions. The dark bands range in color from black to amber.

Paleomagnetism/Rock Magnetism. NRM intensities for the dark specimens are an order of magnitude stronger (average 8.0×10^{-4} A/m) than the NRM of lighter specimens (average 5.0×10^{-5} A/m). The magnetization in lighter-colored specimens did not decay in a stable manner during demagnetization, presumably because of their weak magnetization. However, the dark, hydrocarbon-saturated specimens yield stable, consistent decay in a south-southeasterly and shallow direction during both AF and thermal demagnetization (Figures 3a and 3b). Equal area projections of the specimen directions from all sites are shown on Figure 4. Site and mean statistics are listed in Table 1. The mean direction is declination=160°, inclination=3°, and the corresponding pole position is 113°E, 49°N. When plotted on the Apparent Polar Wander Path (APWP) for stable North America [Irving and Irving, 1982], this pole position suggests the acquisition of remanence in the Permian (Figure 5).

Specimens from one isolated speleothem (site BQ4) contain a more complex magnetization. Demagnetization trajectories for most specimens from site BQ4 change from a southeasterly direction to a south-southeasterly direction during progressive treatments. This curved demagnetization path suggests the simultaneous removal of two components. The magnetization in a few specimens, however, displayed linear decay in an east-southeasterly and shallow direction, and the magnetization in one specimen decayed in a south-southeasterly and shallow direction. The data were analyzed using remagnetization circles, and the circles converged at a southeasterly and shallow direction [Elmore et al., 1988]. The trends of the demagnetization trajectories suggest that the unresolved component is south-southeasterly and shallow, and it may correspond to the Permian direction found in the other speleothems at location BQ. As discussed by Elmore et al. [1988], the easterly direction for the speleothems is interpreted to be anomalous due to structural rotation. The speleothem at site BQ4 apparently acquired part of its total magnetization (the easterly and shallow component) before rotation, and part (the unresolved but probably south-southeasterly and shallow component) presumably after rotation.

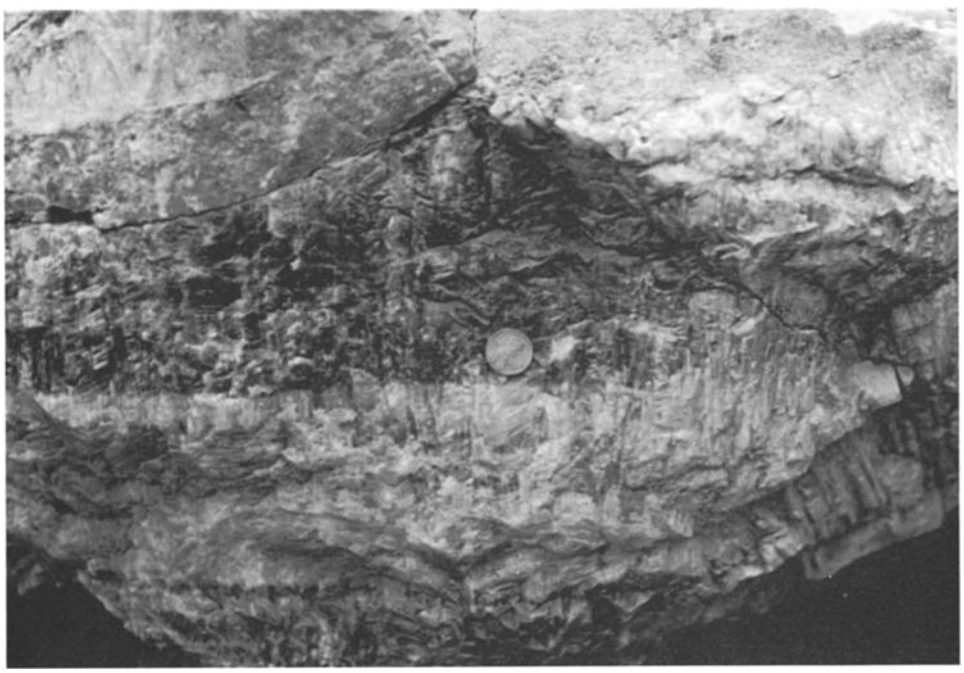


Fig. 2. Light and dark speleothem deposit. The dark speleothem (above horizontal line at base of penny) contains abundant hydrocarbon inclusions. The speleothem below the penny is light and does not contain significant hydrocarbons. The penny (19 mm) is shown for scale.

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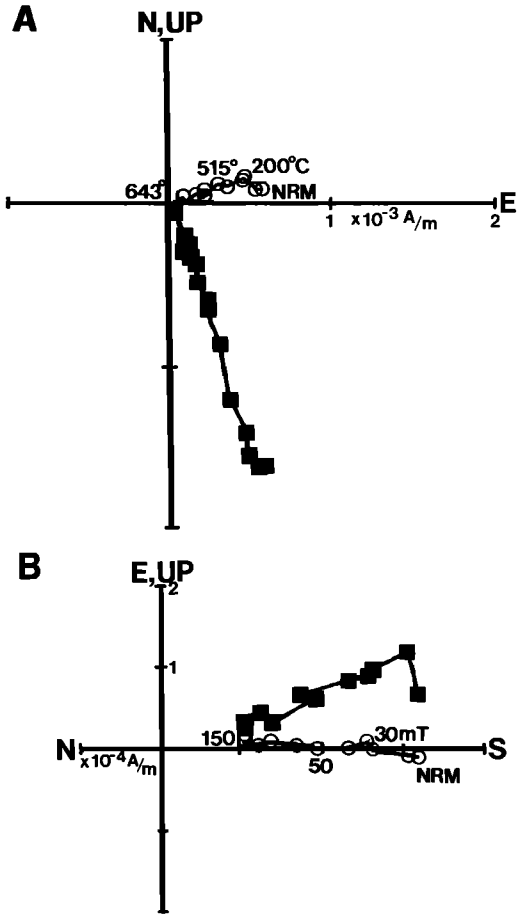


Fig. 3. Examples of orthogonal projections for both (a) thermal and (b) AF decay of dark calcites. Squares indicate the horizontal component; circles indicate the vertical component.

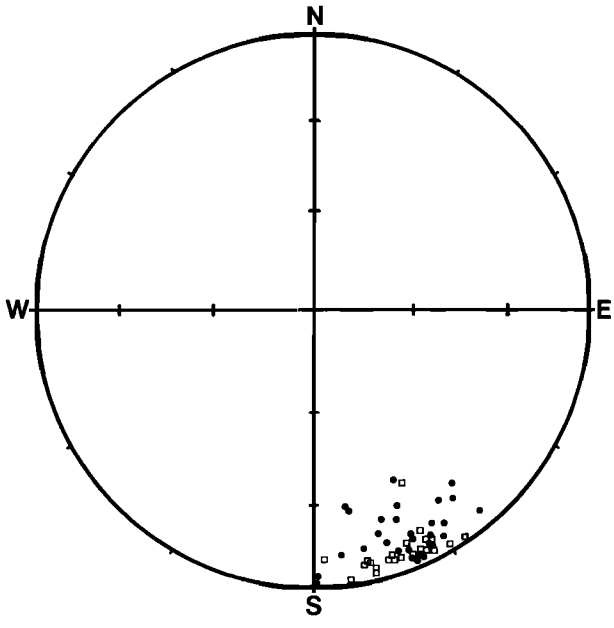


Fig. 4. Equal area projection of specimen directions from the dark speleothems. Open squares indicate a negative inclination, and solid circles indicate a positive inclination.

Maximum unblocking temperatures for all speleothems ranged from 550°C to 643°C (Figure 6a). In many specimens the magnetization was removed by 580°C, which suggests the presence of magnetite. Specimens exhibited significant decay with AF demagnetization, but some were not totally demagnetized (Figure 3b).

Isothermal remanent acquisition experiments for the dark, hydrocarbon-saturated specimens show a steep initial rise in IRM by 0.3 T followed by a flattening off of the slope, although a gradual rise above 0.3 T is also apparent (Figure 6b). This suggests that a low coercivity phase, such as magnetite, is dominant, while a higher coercivity phase, such as hematite, is also present. Thermal decay of the IRM indicates the magnetization is almost entirely removed by 580°C with minor amounts of magnetization remaining up to 650°C (Figure 6c).

After the application of an external field of 0.8 T, some dark specimens were subjected to an antiparallel field of 0.2 T and then thermally demagnetized (Figure 6d). The decay to a minimum of approximately 555°C probably represents the point where the remaining antiparallel 0.20-T and 0.80-T IRMs are equal and almost cancel each other out. The complete decay of magnetite is likely represented by the increase in intensity at approximately 570°C. At temperatures above 580°C only a very small amount of a higher-coercivity IRM is present, and it decays by 680°C. These results suggest that the magnetization resides dominantly in magnetite with some contribution from a higher coercivity mineral, probably hematite.

Petrographic Samples. Transmitted and reflected light microscopy of the dark speleothems reveals columnar calcite crystals in which growth bands are defined by abundant hydrocarbon inclusions (Figures 7a-c). The hydrocarbon inclusions commonly define former crystal faces (Figure 7d). The hydrocarbon inclusions along growth planes are generally small (<3 μm), whereas some larger inclusions are found within crystals and as inclusions associated with fractures and crystal boundaries. The hydrocarbon inclusions found along growth bands are primary in origin. In contrast, the light calcites are clean with very few hydrocarbon inclusions present. The presence of primary hydrocarbon inclusions along growth planes in the dark speleothems suggests that hydrocarbons seeped into caves during the precipitation of the speleothems and were trapped in the growing calcite crystals.

Petrographic study also reveals the presence of several opaque minerals in the calcites, including pyrite and hematite. The most abundant opaque mineral is pyrite which is thought to be syngenetic in origin. In some cases, hematite has apparently replaced pyrite around the margins of some crystals (Figure 7e).

SEM examination of the magnetic extracts from the dark calcites revealed spherical (Figure 7f), botryoidal, and other authigenic forms which, based on energy dispersive analysis, contain iron as the only or major detectable element. The X-ray diffraction (XRD) patterns of extracts from the dark calcites indicate the presence of both magnetite and hematite. The spherical and botryoidal forms are interpreted as authigenic

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TABLE 1. Summary of Paleomagnetic Results for Speleothem Sites From Location BQ

Site	N/No	Decl/Incl deg.	k	α95	Pole Position, °E/°N
BQ1	19/19	155/8	27	7	118/45
BQ2	5/6	165/4	19	18	105/50
BQ6	16/16	159/-4	119	3	116/52
BQ7	2/4	174/-2	-	-	93/56
BQ8	4/4	166/12	38	15	101/47
BQ9	10/10	156/11	53	7	116/44
All specimens	56/59	160/3	17	5	113/49

Decl is declination, and Incl is inclination; k is Fisher's [1953] precision parameter; alpha 95 is radius of cone of 95% confidence; N/No is ratio of specimens used in statistical calculations to specimens analyzed. The semiaxes of the oval of 95% confidence around the mean pole position are dp=2.4 (error in colatitude) and dm=4 (error in declination).

magnetite. These forms are similar to authigenic magnetite reported in the literature by other workers [e.g., McCabe et al., 1983]. Examinations of polished grain mounts in reflected light are consistent with the interpretation that the spheres are magnetite. Platy aggregates and needlelike phases where iron is the only or major detectable element based on EDA analysis are also present. Most of these phases are probably authigenic hematite. Framboidal phases with iron as the only detectable element are interpreted to be pyrite framboids which have been replaced by hematite or perhaps magnetite. A few platy to tabular forms contain sulfur based on EDA analysis, and could be pyrrhotite. The rock magnetic and X-ray diffraction studies provided no conclusive evidence for pyrrhotite, and if it is present, the abundance is low.

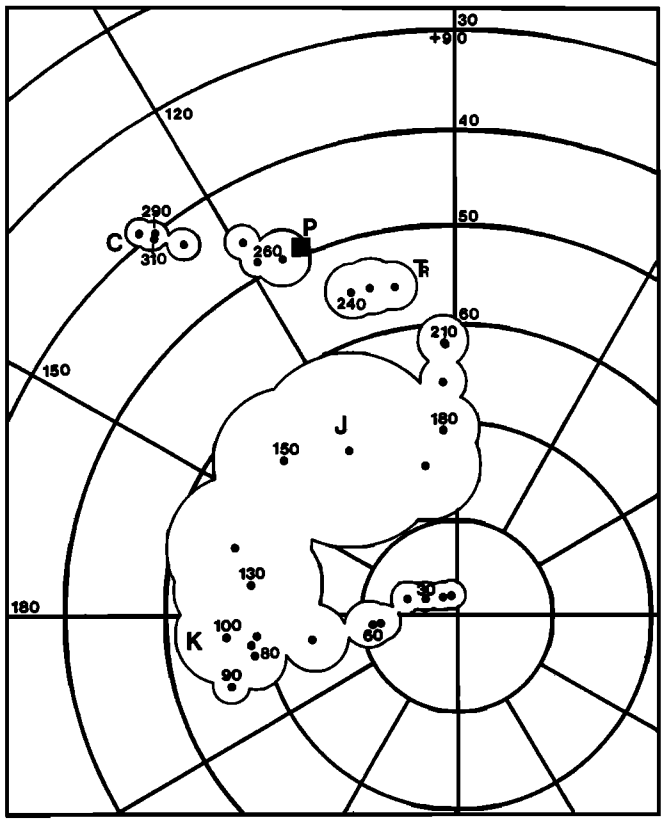


Fig. 5. Apparent Polar Wander Path (APWP) for stable North America [after Irving and Irving, 1982] with the apparent pole position (square) for the dark speleothems. The pole position suggests acquisition of remanence in the Permian. Circles around the 10-Ma poles of the APWP correspond to the standard errors.

Geochemistry. Chromatographic analyses of the C_{15+} aliphatic fraction isolated from the speleothems indicate that biodegradation is slight (Figure 8a). Because the light C_1-C_4 hydrocarbon fraction was not analyzed, we cannot evaluate the level of degradation in the lower molecular weight saturated hydrocarbons. It is reasonable, however, that the lower molecular weights have undergone some degradation due to water washing [e.g., Lafargue and Barker, 1988] or perhaps microbial activity [e.g., Sassen et al., 1989]. It is clear, however, that the oil has not undergone any extensive aerobic biodegradation, which is the case for solid bitumen (e.g., Figure 8b) studied by Elmore et al. [1987] and McCabe et al. [1987]. It is likely that once the hydrocarbons were entrapped in the fluid inclusions, they were isolated from further degradation. The results of pyrolysis experiments (thermal decrepitation) reveal more than one mode of occurrence of hydrocarbon inclusions in the speleothem crystals, which supports the petrographic analysis. Analysis by GC and GC/mass spectrometry suggests that the most likely source of the oil in the speleothems is the Devonian Woodford Formation (M. H. Engel and S. Imbus, personal communication, 1989).

Stable carbon and oxygen isotope values for the light ($\delta^{13}C_{PDB}$, -6.7‰; $\delta^{18}O_{PDB}$, -4.2‰) and dark ($\delta^{13}C_{PDB}$, -6.2‰; $\delta^{18}O_{PDB}$, -4.7‰) are slightly depleted. The stable carbon isotopic value for the hydrocarbons in the speleothems is -30.7‰ (PDB). The isotopic values in the calcites could be slightly depleted due to the incorporation of some isotopically light carbon and oxygen from limited oxidation of hydrocarbons during precipitation of the speleothems. The fact that the calcites are not more depleted supports the interpretation that the hydrocarbons

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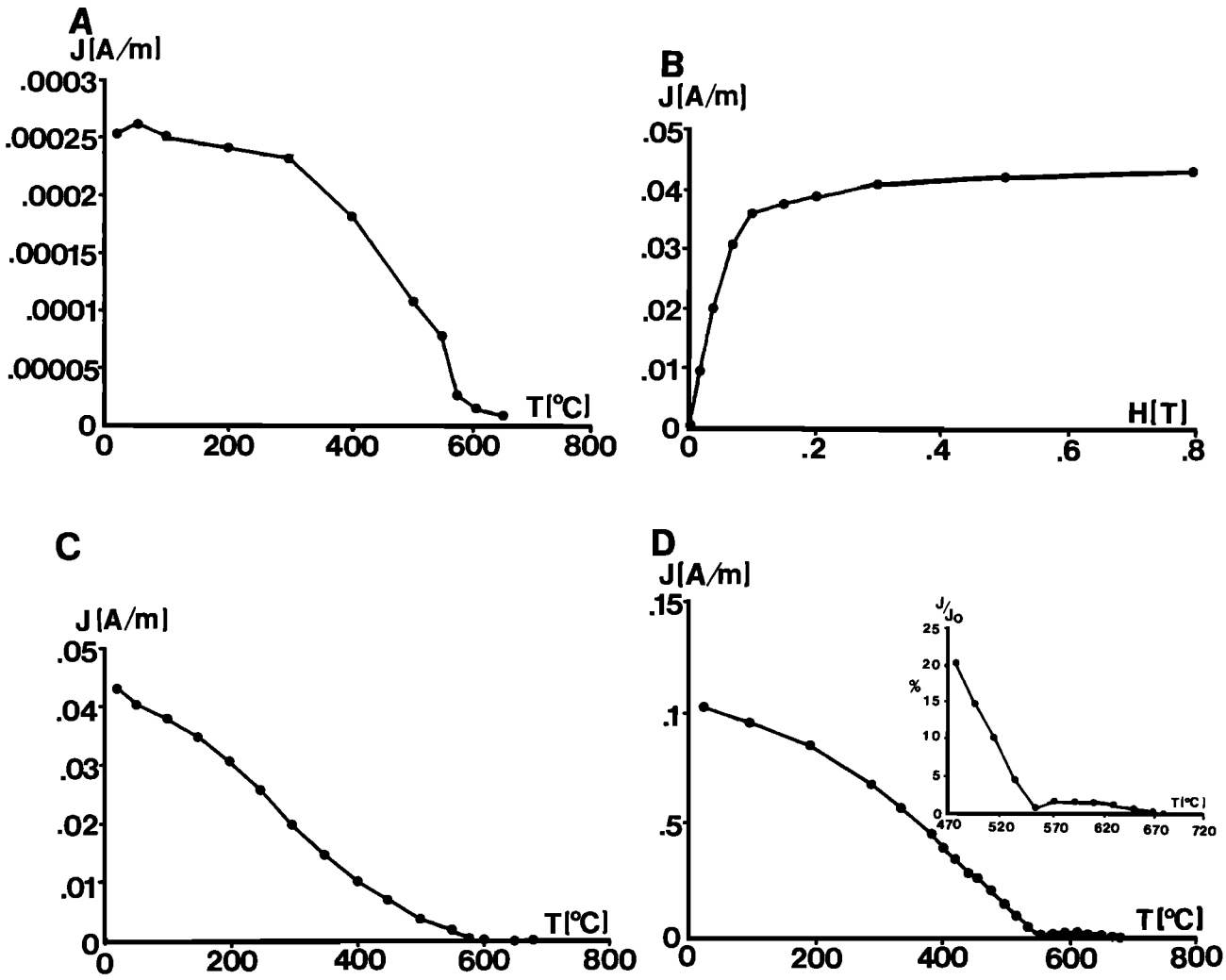


Fig. 6. Magnetic characteristics of the dark calcites from the speleothems: (a) thermal decay of the NRM, (b) representative IRM acquisition curve, (c) thermal decay of the IRM (same specimen as in Figure 6b), and (d) thermal decay of a reverse field IRM, where the insert illustrates decay (in percent of total) at high temperatures.

were not extensively degraded when trapped in the fluid inclusions. The values in the calcites, however, are consistent with a shallow depth of formation from fresh water [Allan and Matthews, 1977]. The similarity of $\delta^{13}C$ and $\delta^{18}O$ values for the adjacent dark and light calcites also favors a common origin.

Interpretation. The magnetization in the speleothems was acquired in the Permian, predominantly resides in authigenic magnetite, and is interpreted as a CRM. The fact that the magnetization in some specimens exhibited linear decay up to 643°C suggests that some of the magnetization may reside in hematite or possibly maghemite. The results of the rock magnetic experiments and petrographic studies, as well as the incomplete AF demagnetization in some cases (Figure 3b), suggest that a high coercivity mineral, hematite, is present and carries some of the magnetization. The evidence for the presence of maghemite is inconclusive. The demagnetization results and the results of the rock magnetic experiments, however, suggest that most of the magnetization resides in magnetite. The fact that the cave deposits formed at relatively shallow depths and that the Arbuckle Group speleothems were never deeply buried eliminates the possibility of a thermoviscous origin for the magnetization. The stable isotope values for the speleothems are also consistent with a shallow depth of formation [Allan and Matthews, 1977].

The results from the light and dark speleothems suggest that there is a relationship between hydrocarbons and authigenic magnetite; the stronger magnetization invariably resides in the dark calcites, where the hydrocarbons are concentrated. This relationship between intensity of magnetization and abundance of hydrocarbons suggests that hydrocarbons caused precipitation of authigenic magnetite and acquisition of the associated CRM. The fact that the time of remanence acquisition is consistent with the time of hydrocarbon migration and

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Fig. 7. Petrographic characteristics of the hydrocarbon-saturated speleotherms.

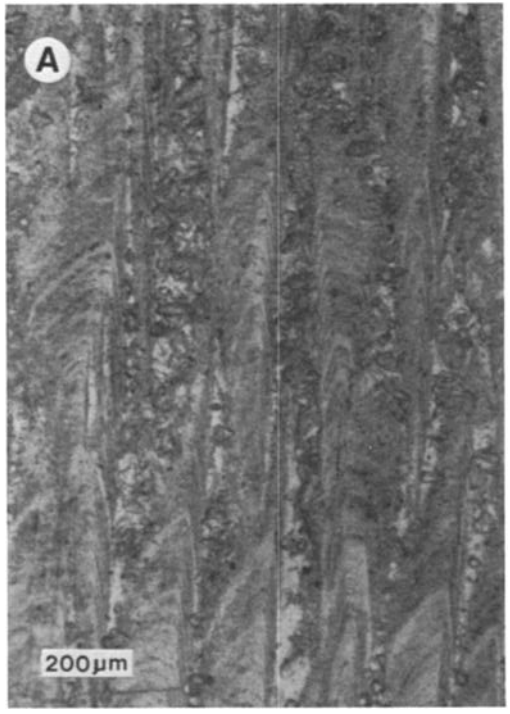


Fig. 7a. The speleothems are composed of columnar crystals which contain growth bands delineated by primary hydrocarbon-filled inclusions (plane light).

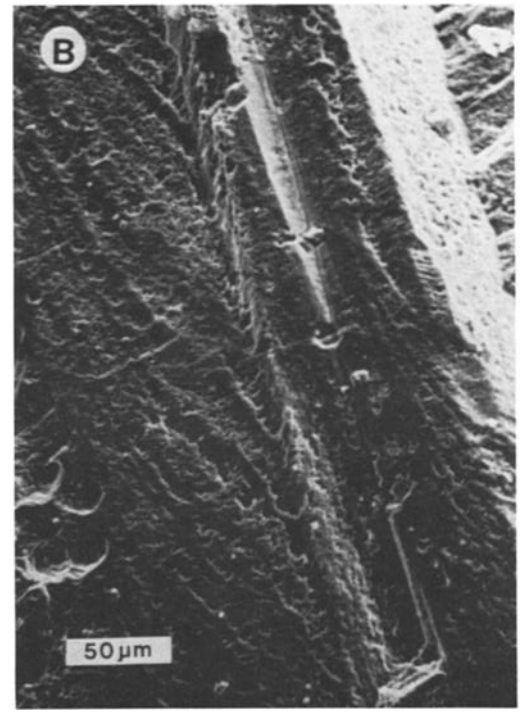


Fig. 7b. SEM photomicrograph of columnar calcite with linear inclusion bands.

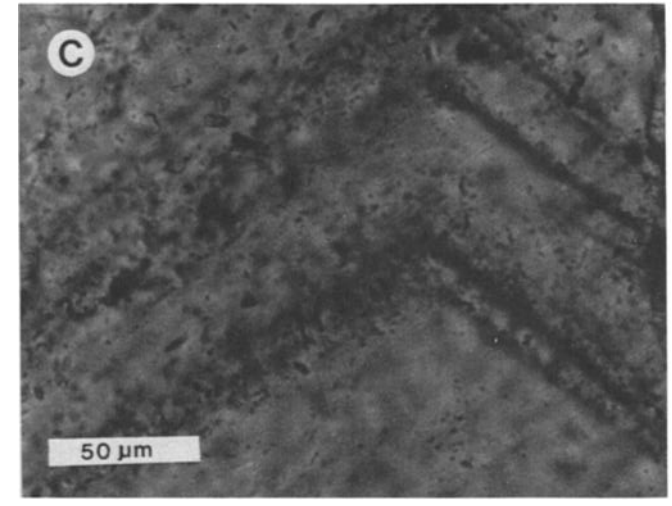


Fig. 7c. Close up of growth band composed of primary hydrocarbon-filled inclusions (plane light).

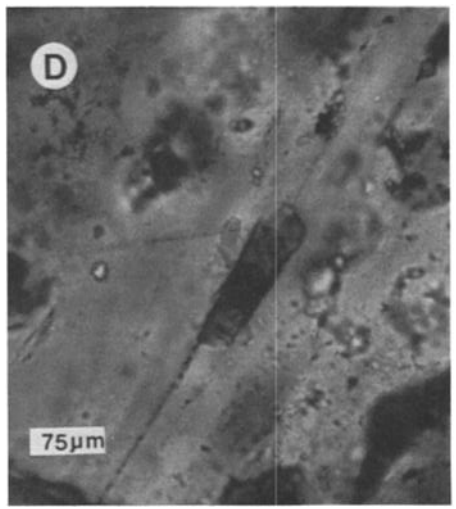


Fig. 7d. Close up of hydrocarbon-filled inclusion in speleothem calcite (plane light).

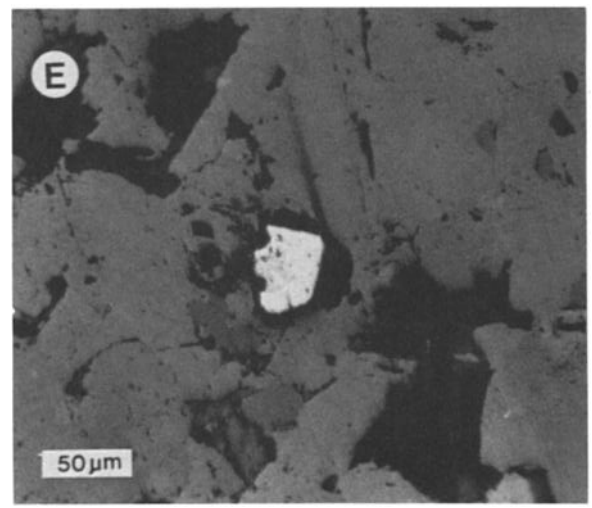


Fig. 7e. Reflected light photomicrograph of hematite replacing pyrite. Pyrite is in the center of the grain.

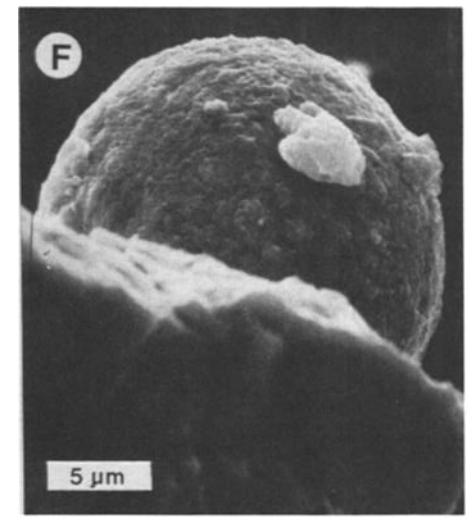


Fig. 7f. SEM photomicrograph of a sphere interpreted to be augthigenic magnetite.

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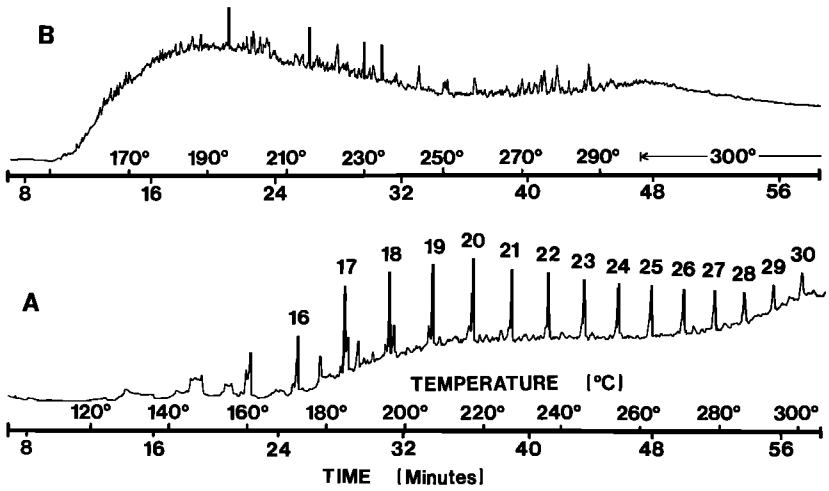


Fig. 8. Chromatograms of the saturate fraction (a) of the hydrocarbon-saturated speleothems and (b) of solid bitumen. The solid bitumen is extensively degraded (note attenuation of peaks), whereas the C 15+ fraction of the speleothem oil is only slightly degraded. Analysis of the speleothem oil was performed with a Varian 330 gas chromatograph (GC) equipped with a flame ionization detector (FID) and a 30m x 0.25 mm ID J&W bonded-phase column coated with a 1.0 m-film of DB-5. The injector and detector temperatures were 300 C. Sample injection was done in the splitless mode. After 0.5 min, the split was adjusted 50:1. The carrier gas (He) flow rate was 2 mL/min. The oven temperature program was 40 C isothermal for 1 min, and then 40 to 100 C at 10 C/min and 100 to 300 C at 4 C/min. The run was finished isothermal at 300 C. The solid bitumen analysis was performed with a Hewlitt Packard 5890A GC equipped with an FID detector and a 30m x 0.25mm ID J&W bonded-phase column coated with a 1.0 um film of DB-5. The injector and detector temperatures were 300 C. Sample injector was done in the splitless mode. The carrier gas (He) flow rate was 1.0 mL/min. The oven temperature program was 40 C isothermal for 4 min, and then 40 to 150 C at 20 C/min and 150 to 300 C at 4 C/min. The run was finished isothermal at 300 C. Modified from Figure 3, Elmore et al. (1987). Reprinted by permission from Nature, vol. 325, pp. 429. Copyright(s) 1987, MacMillon Magazines Ltd.

formation of the speleothems also provides evidence that hydrocarbons can cause acquisition of a stable secondary magnetization residing in magnetite. The origin of the hematite in the speleothems is problematic. It could have formed by direct precipitation due to changing chemical conditions. Alternatively, the hematite could have replaced magnetite soon after speleothem formation.

Host Limestones: Kindblade Formation

Paleomagnetic results from the host limestones are presented by Elmore et al. [1988]. The mean in situ direction is declination=131°, inclination=23° (k=38, $\alpha 95$=12°, n=5 sites). Determining the time of remanence acquisition in these limestones is not straightforward because the results are not constrained by a fold test, and the rocks may have been rotated as a result of left-lateral wrench faulting [Elmore et al., 1988]. Elmore et al. [1988] report that other Arbuckle Group carbonates in the Limestone Hills fail a fold test, which suggests the magnetization is post-Pennsylvanian in age. This, and the relatively shallow inclinations for the in situ directions, lead to the suggestion that the carbonates, including the host limestones for the speleothems, acquired their magnetization in the Permian [Elmore et al., 1988].

Demagnetization and rock magnetic results for the host limestones suggest that the magnetization is dominated by magnetite. The magnetization in specimens of the limestones decayed during both AF and thermal treatment. Maximum unblocking temperatures for the specimens generally ranged between 500 and 575°C (Figure 9a). Rock magnetic experiments indicate the presence of a low-coercivity phase in all specimens examined, as an abrupt rise in IRM acquisition by 0.3 T is observed (Figure 9b). Some of the curves gradually rise above 0.3 T, which also suggests the presence of a higher-coercivity phase such as hematite. IRM decay indicates that at least 95% of the IRM is removed by 580°C (Figure 9c), which suggests the presence of magnetite. Decay of an antiparallel field IRM also suggests that magnetite is present (Figure 9d).

The magnetization in the limestones is interpreted to be secondary and predominantly resides in magnetite. A thermoviscous origin for the magnetization can probably be eliminated because of low burial depths and temperatures for the rocks in the area [Cardott and Lambert, 1985]. The magnetization is, therefore, probably

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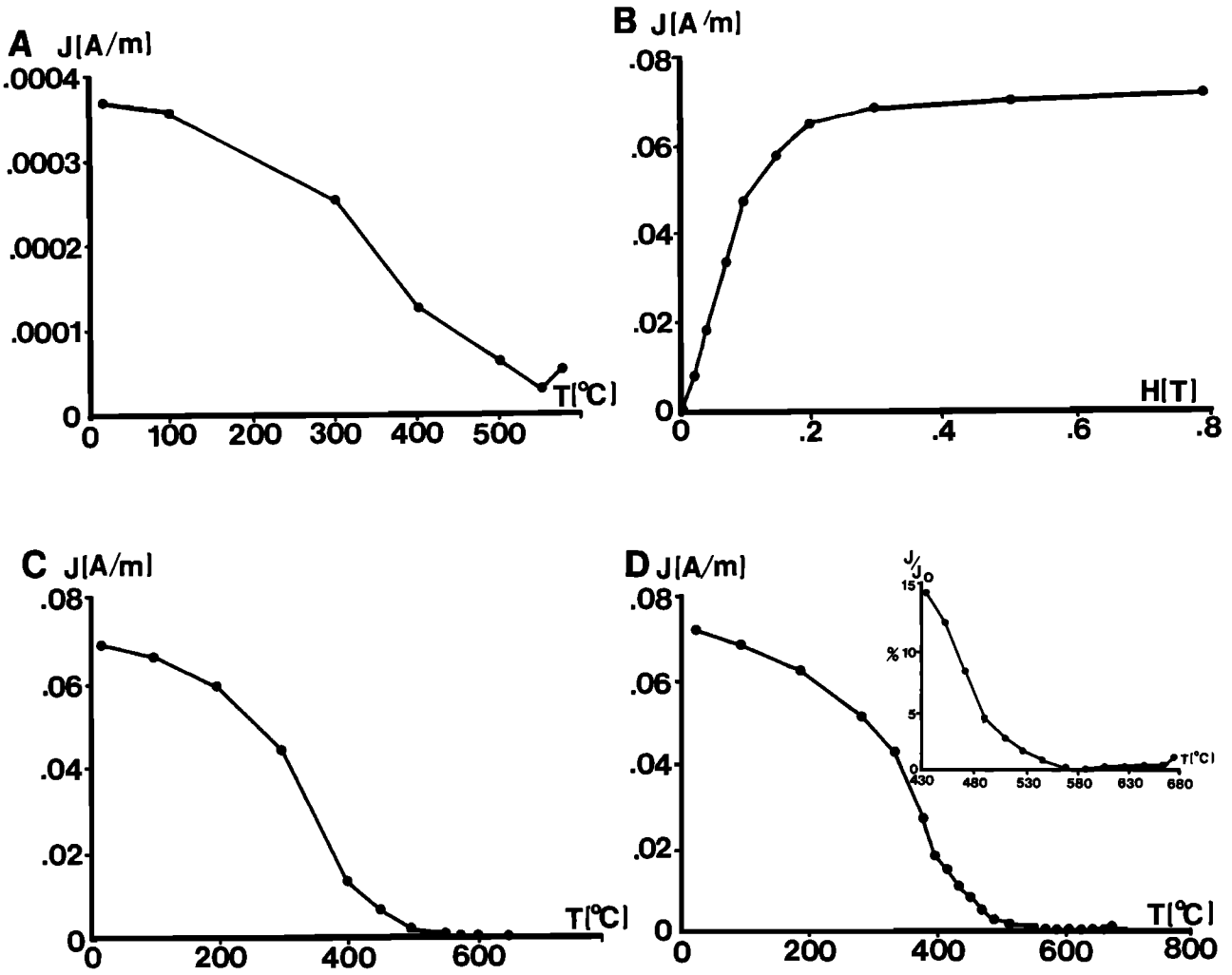


Fig. 9. Magnetic characteristics of the Ordovician limestones wich are the host rocks for the speleothems: (a) thermal decay of NRM, (b) representative IRM acquisition curve, (c) thermal decay of IRM (same specimen as in Figure 9b), and (d) decay of a reverse field IRM with an insert illustrating decay (in percent of total) at high temperatures.

a CRM and may have been acquired by a mechanism similar to that in the speleothems. The results are consistent with the interpretation that the hydrocarbons created the chemical conditions necessary for authigenic magnetite to precipitate. Where similar limestones do not contain hydrocarbons, they possess a CRM residing in authigenic hematite [e.g., Cochran and Elmore, 1987].

Discussion

The relationship between intensity of magnetization and abundance of hydrocarbons in the speleothems suggests that the chemical environment created by the hydrocarbons caused precipitation of authigenic magnetite. The fact that the time of remanence acquisition in the speleothems is consistent with the time of hydrocarbon migration and age of the speleothems also provides evidence that hydrocarbons can cause acquisition of a stable secondary remanence. This not only has implications for understanding remagnetization mechanisms, but also suggests that it may be possible to date or constrain the timing of hydrocarbon migration by isolating the magnetization carried in authigenic magnetite.

Remagnetized, magnetite-bearing limestones are common [e.g., McCabe et al., 1983]. The fact that hydrocarbons can cause precipitation of authigenic magnetite and acquisition of a stable remanence is important for understanding remagnetization mechanisms in these limestones. Undoubtedly, some of these limestones contain thermoviscous magnetizations [e.g., Kent, 1985], whereas other limestones contain CRMs. Some of these CRMs may reside in authigenic magnetite that precipitated due to hydrocarbon migration.

Although the results from the speleothems and solid bitumen provide empirical evidence for a relationship between hydrocarbons and authigenic magnetite, the geochemical mechanism for the formation of the magnetite can only be speculated on at this time. Biodegradation of hydrocarbons has been suggested as causing the precipitation

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of some authigenic minerals such as pyrite and pyrrhotite [Sassen, 1980]. The presence of authigenic magnetite in solid bitumen that is degraded suggests that microbial attack of hydrocarbons is one possible diagenetic pathway that could lead to precipitation of magnetite [e.g., Elmore et al., 1987; McCabe et al., 1987]. Most hydrocarbons at the surface of the Earth, however, are extensively degraded. The authigenic magnetite in the speleothems may have formed due to biodegradation, but an origin related to other chemical conditions created by hydrocarbons cannot be ruled out. The fact that the C_{15+} hydrocarbons in the speleothems are only slightly degraded also suggests that biodegradation may not be the only pathway for the formation of diagenetic magnetite. Berner [1964], for example, showed that formation of diagenetic magnetite is thermodynamically possible for several reactions (e.g., pyrite-magnetite). Future work may provide new information on the specific chemical conditions created by the hydrocarbons that cause magnetite authigenesis.

Although it is premature to suggest a specific chemical reaction for precipitation of magnetite in the speleothems, there are several sources for the iron that is needed. The hydrocarbons themselves may have provided the iron, as iron is found in oil [Ellrich et al., 1985]. Fluids associated with hydrocarbons also contain iron [Kharaka et al., 1985]. Iron dissolved from the rocks that the hydrocarbons or associated fluids migrated through could also have been a source [e.g., Donovan et al., 1979].

Fluids other than hydrocarbons may also cause magnetite precipitation. For example, Hornafius [1984] suggests that meteoric waters may cause oxidation of pyrite to magnetite in the Monterey Formation. Hart and Fuller [1988] report a CRM residing in authigenic magnetite in the Monterey Formation and associate the magnetite with fluids that caused late stage dolomite diagenesis. Oxidation of siderite by fluids can also produce authigenic magnetite [e.g., Ellwood et al., 1988]. The release of iron into pore waters during the transformation of smectite to illite during burial [e.g., Boles and Frank, 1979] could also produce a fluid that may be capable of precipitating magnetite.

Oliver [1986] suggested that fluids, including hydrocarbons, were expelled from orogenic belts and moved through permeable units onto the craton. He suggested that the distribution of Mississippi Valley-type lead-zinc deposits and hydrocarbon reservoirs, as well as the widespread late Paleozoic remagnetizations in North America, may be a consequence of regional scale fluid migration. Although the remanence in the speleothems is not due to remagnetization, the origin of this magnetization is probably related to fluids (hydrocarbons) expelled during deformation. The speleothems are found in a deformed limestone terrain that is adjacent to and faulted over the Anadarko Basin [Donovan, 1986]. Generation of hydrocarbons from the Woodford Shale in the Anadarko Basin probably occurred in the Pennsylvanian before and during deformation [Cardott and Lambert, 1985]. Some of these hydrocarbons migrated into the overlying limestones and caves along conduits produced during deformation and caused acquisition of the magnetization in the speleothems. Other studies have also related magnetizations to fluids generated during deformation [e.g., Miller and Kent, 1988].

The establishment of a relation between hydrocarbons and magnetite has obvious implications for the orogenic fluid model. It is possible that paleomagnetic analysis may be used to test the tectonic fluid model by a careful study of magnetite-bearing units. If the model is correct, then the timing of remagnetization should be temporally related to major tectonic events and spatially related to deformed zones.

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